

(a)searchEvents.jsp

```
<%@ page contentType="text/html" pageEncoding="UTF-8" %>
<!DOCTYPE html>
<html>
<head>
  <title>EventSphere - Search Events</title>
</head>
<body>
  <h2>EventSphere Event Booking System</h2>

  <form action="searchEventServlet" method="get">
    <label>Budget:</label>
    <input type="text" name="budget" />
    <br /><br />

    <label>Venue:</label>
    <select name="venue">
      <option value="">-- Select Venue --</option>
      <option value="Convention Center">Convention Center</option>
      <option value="Harbour Park">Harbour Park</option>
      <option value="City Gallery">City Gallery</option>
    </select>
    <br /><br />

    <input type="submit" value="FIND EVENTS" />
  </form>
</body>
</html>
```

(b)EventBean.java

```
package com.bean;
```

```
public class EventBean {  
    private int eventID;  
    private String eventName;  
    private String eventDate;  
    private String venue;  
    private int ticketPrice;  
    private int seatsAvailable;  
  
    public EventBean() {  
    }  
  
    public int getEventID() {  
        return eventID;  
    }  
  
    public void setEventID(int eventID) {  
        this.eventID = eventID;  
    }  
  
    public String getEventName() {  
        return eventName;  
    }  
  
    public void setEventName(String eventName) {  
        this.eventName = eventName;  
    }  
  
    public String getEventDate() {  
        return eventDate;  
    }  
  
    public void setEventDate(String eventDate) {  
        this.eventDate = eventDate;  
    }  
  
    public String getVenue() {
```

```
        return venue;
    }

    public void setVenue(String venue) {
        this.venue = venue;
    }

    public int getTicketPrice() {
        return ticketPrice;
    }

    public void setTicketPrice(int ticketPrice) {
        this.ticketPrice = ticketPrice;
    }

    public int getSeatsAvailable() {
        return seatsAvailable;
    }

    public void setSeatsAvailable(int seatsAvailable) {
        this.seatsAvailable = seatsAvailable;
    }
}
```

(c)EventDB.java

```
package com.db;
```

```
import com.bean.EventBean;  
import java.sql.Connection;  
import java.sql.DriverManager;  
import java.sql.ResultSet;  
import java.sql.Statement;  
import java.util.ArrayList;
```

```
public class EventDB {
```

```
    private String dbUrl;  
    private String dbUser;  
    private String dbPassword;
```

```
    public EventDB(String dbUrl, String dbUser, String dbPassword) {  
        this.dbUrl = dbUrl;  
        this.dbUser = dbUser;  
        this.dbPassword = dbPassword;  
    }
```

```
    public Connection getConnection() throws Exception {  
        Class.forName("com.mysql.jdbc.Driver");  
        return DriverManager.getConnection(dbUrl, dbUser, dbPassword);  
    }
```

```
    public ArrayList<EventBean> getEvents(String venue) throws Exception {  
        ArrayList<EventBean> events = new ArrayList<EventBean>();  
        String sql = "SELECT * FROM eventTable WHERE venue = '" + venue + "'";
```

```
        Connection conn = getConnection();  
        Statement stmt = conn.createStatement();  
        ResultSet rs = stmt.executeQuery(sql);
```

```
        while (rs.next()) {  
            EventBean e = new EventBean();  
            e.setEventID(rs.getInt("eventID"));  
            e.setEventName(rs.getString("eventName"));  
            e.setEventDate(rs.getString("eventDate"));
```

```
        e.setVenue(rs.getString("venue"));
        e.setTicketPrice(rs.getInt("ticketPrice"));
        e.setSeatsAvailable(rs.getInt("seatsAvailable"));
        events.add(e);
    }

    rs.close();
    stmt.close();
    conn.close();

    return events;
}
}
```

(d)ListEventServlet.java

```
package com.servlet;
```

```
import com.bean.EventBean;
import com.db.EventDB;
import java.io.IOException;
import java.util.ArrayList;
import javax.servlet.RequestDispatcher;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
```

```
@WebServlet("/searchEventServlet")
public class ListEventServlet extends HttpServlet {
```

```
    private EventDB eventDB;
```

```
    @Override
```

```
    public void init() throws ServletException {
        eventDB = new EventDB(
            "jdbc:mysql://localhost:3306/EVENTSDB",
            "root",
            ""
        );
    }
}
```

```
    @Override
```

```
    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
        throws ServletException, IOException {
        processRequest(request, response);
    }
}
```

```
    @Override
```

```
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
        throws ServletException, IOException {
        processRequest(request, response);
    }
}
```

```

    }

    private void processRequest(HttpServletRequest request,
    HttpServletResponse response)
        throws ServletException, IOException {
        String venue = request.getParameter("venue");
        String budgetStr = request.getParameter("budget");

        if (venue == null || venue.trim().isEmpty() || budgetStr == null ||
        budgetStr.trim().isEmpty()) {
            response.sendError(HttpServletResponse.SC_BAD_REQUEST, "Missing
            budget or venue");
            return;
        }

        try {
            int budget = Integer.parseInt(budgetStr);
            ArrayList<EventBean> allEvents = eventDB.getEvents(venue);
            ArrayList<EventBean> filteredEvents = new ArrayList<EventBean>();

            for (EventBean e : allEvents) {
                if (e.getTicketPrice() <= budget) {
                    filteredEvents.add(e);
                }
            }

            request.setAttribute("eventList", filteredEvents);
            request.getRequestDispatcher("/list.jsp").forward(request, response);
        } catch (Exception ex) {
            throw new ServletException(ex);
        }
    }
}

```

(e)List.jsp

```
<%@ page contentType="text/html" pageEncoding="UTF-8" %>
<%@ page import="java.util.ArrayList, com.bean.EventBean" %>
<!DOCTYPE html>
<html>
<head>
    <title>Event List</title>
</head>
<body>
    <jsp:useBean id="eventList" class="java.util.ArrayList" scope="request" />
    <%
        ArrayList<EventBean> events = (ArrayList<EventBean>)
request.getAttribute("eventList");
        if (events == null) {
            events = new ArrayList<EventBean>();
        }

        int totalSeats = 0;
    %>

    <h2>Matching Events</h2>

    <table border="1" cellpadding="6" cellspacing="0">
        <tr>
            <th>Event ID</th>
            <th>Event Name</th>
            <th>Event Date</th>
            <th>Venue</th>
            <th>Ticket Price</th>
            <th>Seats Available</th>
        </tr>

        <%
            for (EventBean e : events) {
                totalSeats += e.getSeatsAvailable();
            %>
        <tr>
            <td><%= e.getEventID() %></td>
```

```
<td><%= e.getEventName() %></td>
<td><%= e.getEventDate() %></td>
<td><%= e.getVenue() %></td>
<td><%= e.getTicketPrice() %></td>
<td><%= e.getSeatsAvailable() %></td>
</tr>
<%
    }
%>
</table>

<p>Total seats available: <%= totalSeats %></p>
</body>
</html>
```

(f) Understanding on system architecture

Naming server in microservices architecture is a service registry. Each microservice registers its instance address and service name with it, and other services look up the logical name instead of hard-coding IP addresses or ports. This makes services easier to discover, scale, load-balance, and replace when instances change.